Supplementary Material

for "FACEO-R: Functionally Anchored Overlapping Community Detection for Robust Pathway Module Discovery"

Supplementary Figures

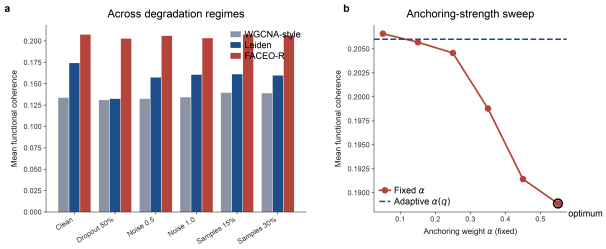


Fig. S1. The advantage is present with and without degradation, and the anchoring strength has a narrow optimum. (a) Mean functional coherence under clean data and five degradation regimes; FACEO-R exceeds both baselines in every condition. (b) Coherence as a function of a fixed anchoring weight α (solid); performance improves monotonically as anchoring strengthens, the originally proposed adaptive schedule under-anchored (dashed, below every fixed setting in $[0.05, 0.35]$), and the retuned schedule matches the best fixed weight.

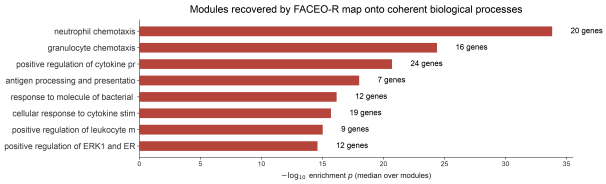


Fig. S2. Recovered modules map onto coherent, specific biological processes. Median enrichment significance of the top Gene Ontology term per module, with the input pathway excluded from consideration; annotations show the median number of module genes in each term. Twenty-two distinct processes were recovered across four cohorts.

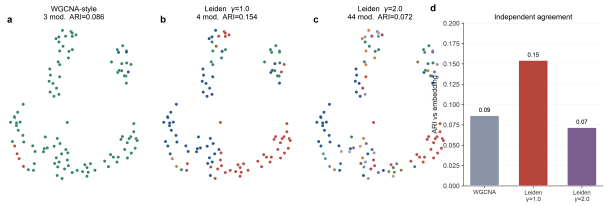


Fig. S3. Self-supervised embedding of the apoptotic subnetwork, projected by UMAP. Nodes are coloured by partition membership for (a) WGCNA-style hierarchical cutting, (b) Leiden $\gamma = 1.0$ and (c) Leiden $\gamma = 2.0$. (d) Adjusted Rand index between each partition and the partition derived from the embedding alone, with no community labels used during training. The embedding reproduces the Leiden $\gamma = 1.0$ partition several-fold better than the hierarchical one.